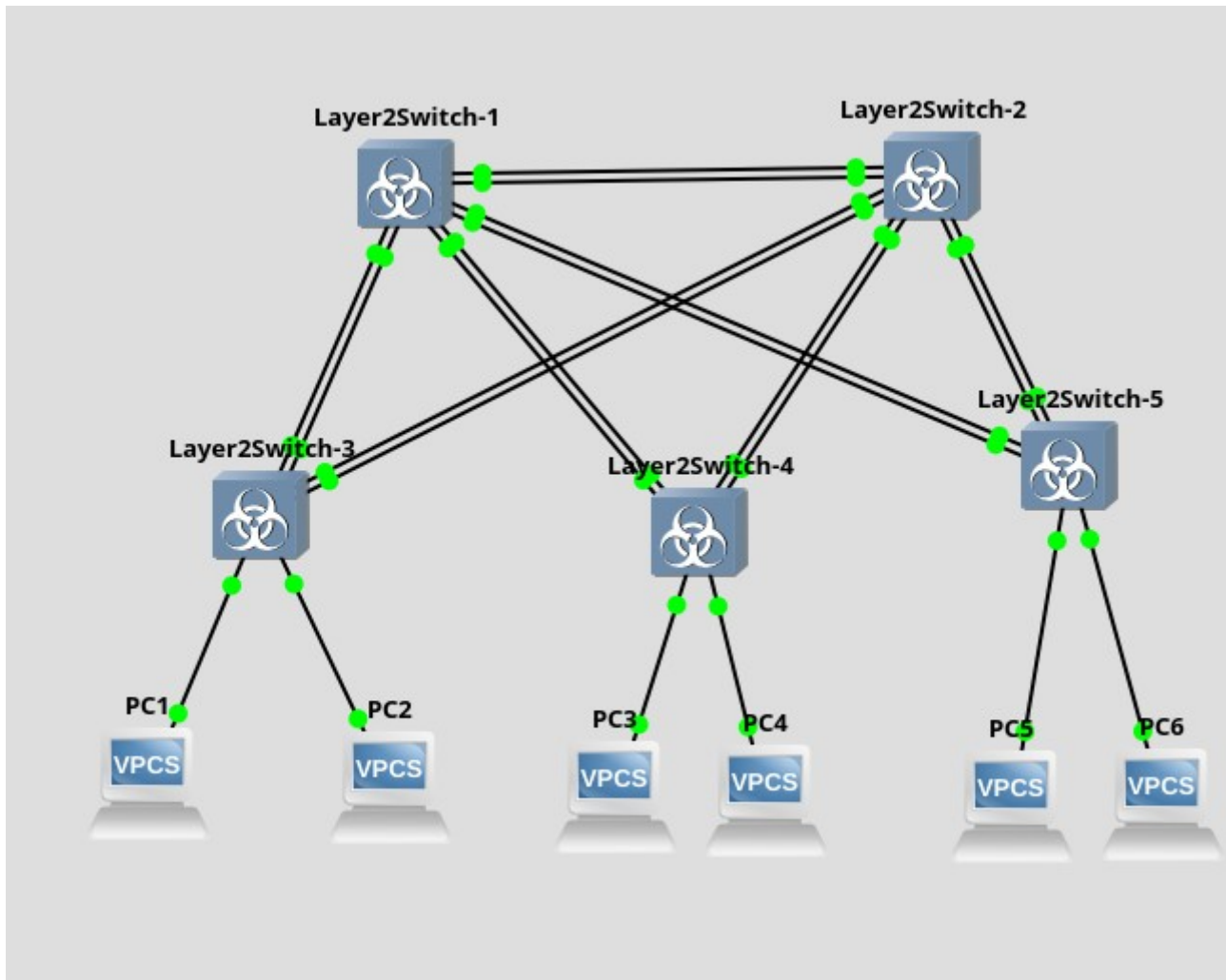


1)



Для коммутаторов без прямого соединения с ПК на все порты устанавливаем VLAN (20 с тегом, а 333 без тега native)

```
Layer2Switch-1
vIOS-L2-01>enable
vIOS-L2-01#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
vIOS-L2-01(config)#no ip domain-lookup
vIOS-L2-01(config)#vlan 20
vIOS-L2-01(config-vlan)#name VLAN20
vIOS-L2-01(config-vlan)#exit
vIOS-L2-01(config)#vlan 333
vIOS-L2-01(config-vlan)#name VLAN333
vIOS-L2-01(config-vlan)#exit
vIOS-L2-01(config)#interface GigabitEthernet0/0
vIOS-L2-01(config-if)#switchport trunk encapsulation dot1q
vIOS-L2-01(config-if)#switchport mode trunk
vIOS-L2-01(config-if)#switchport trunk native vlan 333
vIOS-L2-01(config-if)#switchport trunk allowed vlan 20,333
```

```
vIOS-L2-01(config-if)#no shutdown
vIOS-L2-01(config-if)#exit
vIOS-L2-01(config)#interface GigabitEthernet0/1
vIOS-L2-01(config-if)#switchport trunk encapsulation dot1q
vIOS-L2-01(config-if)#switchport mode trunk
vIOS-L2-01(config-if)#switchport trunk native vlan 333
vIOS-L2-01(config-if)#switchport trunk allowed vlan 20,333
vIOS-L2-01(config-if)#no shutdown
vIOS-L2-01(config-if)#exit
vIOS-L2-01(config)#interface GigabitEthernet0/2
vIOS-L2-01(config-if)#switchport trunk encapsulation dot1q
vIOS-L2-01(config-if)#switchport mode trunk
vIOS-L2-01(config-if)#switchport trunk native vlan 333
vIOS-L2-01(config-if)#switchport trunk allowed vlan 20,333
vIOS-L2-01(config-if)#no shutdown
vIOS-L2-01(config-if)#exit
vIOS-L2-01(config)#interface GigabitEthernet0/3
vIOS-L2-01(config-if)#switchport trunk encapsulation dot1q
vIOS-L2-01(config-if)#switchport mode trunk
vIOS-L2-01(config-if)#switchport trunk native vlan 333
vIOS-L2-01(config-if)#switchport trunk allowed vlan 20,333
vIOS-L2-01(config-if)#no shutdown
vIOS-L2-01(config-if)#exit
vIOS-L2-01(config)#interface GigabitEthernet0/4
vIOS-L2-01(config-if)#switchport trunk encapsulation dot1q
vIOS-L2-01(config-if)#switchport mode trunk
vIOS-L2-01(config-if)#switchport trunk native vlan 333
vIOS-L2-01(config-if)#switchport trunk allowed vlan 20,333
vIOS-L2-01(config-if)#no shutdown
vIOS-L2-01(config-if)#exit
vIOS-L2-01(config)#interface GigabitEthernet0/5
vIOS-L2-01(config-if)#switchport trunk encapsulation dot1q
vIOS-L2-01(config-if)#switchport mode trunk
vIOS-L2-01(config-if)#switchport trunk native vlan 333
vIOS-L2-01(config-if)#switchport trunk allowed vlan 20,333
vIOS-L2-01(config-if)#no shutdown
vIOS-L2-01(config-if)#exit
vIOS-L2-01(config)#interface GigabitEthernet0/6
vIOS-L2-01(config-if)#switchport trunk encapsulation dot1q
vIOS-L2-01(config-if)#switchport mode trunk
vIOS-L2-01(config-if)#switchport trunk native vlan 333
vIOS-L2-01(config-if)#switchport trunk allowed vlan 20,333
vIOS-L2-01(config-if)#no shutdown
vIOS-L2-01(config-if)#exit
vIOS-L2-01(config)#interface GigabitEthernet0/7
vIOS-L2-01(config-if)#switchport trunk encapsulation dot1q
vIOS-L2-01(config-if)#switchport mode trunk
vIOS-L2-01(config-if)#switchport trunk native vlan 333
vIOS-L2-01(config-if)#switchport trunk allowed vlan 20,333
vIOS-L2-01(config-if)#no shutdown
vIOS-L2-01(config-if)#exit
```

Layer2Switch-2

vIOS-L2-01>enable

vIOS-L2-01#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

vIOS-L2-01(config)#no ip domain-lookup

vIOS-L2-01(config)#interface GigabitEthernet0/0

vIOS-L2-01(config-if)#switchport trunk encapsulation dot1q

vIOS-L2-01(config-if)#switchport mode trunk

vIOS-L2-01(config-if)#switchport trunk native vlan 333

vIOS-L2-01(config-if)#switchport trunk allowed vlan 20,333

vIOS-L2-01(config-if)#no shutdown

vIOS-L2-01(config-if)#exit

vIOS-L2-01(config)#interface GigabitEthernet0/1

vIOS-L2-01(config-if)#switchport trunk encapsulation dot1q

vIOS-L2-01(config-if)#switchport mode trunk

vIOS-L2-01(config-if)#switchport trunk native vlan 333

vIOS-L2-01(config-if)#switchport trunk allowed vlan 20,333

vIOS-L2-01(config-if)#no shutdown

vIOS-L2-01(config-if)#exit

vIOS-L2-01(config)#interface GigabitEthernet0/2

vIOS-L2-01(config-if)#switchport trunk encapsulation dot1q

vIOS-L2-01(config-if)#switchport mode trunk

vIOS-L2-01(config-if)#switchport trunk native vlan 333

vIOS-L2-01(config-if)#switchport trunk allowed vlan 20,333

vIOS-L2-01(config-if)#no shutdown

vIOS-L2-01(config-if)#exit

vIOS-L2-01(config)#interface GigabitEthernet0/3

vIOS-L2-01(config-if)#switchport trunk encapsulation dot1q

vIOS-L2-01(config-if)#switchport mode trunk

vIOS-L2-01(config-if)#switchport trunk native vlan 333

vIOS-L2-01(config-if)#switchport trunk allowed vlan 20,333

vIOS-L2-01(config-if)#no shutdown

vIOS-L2-01(config-if)#exit

vIOS-L2-01(config)#interface GigabitEthernet1/0

vIOS-L2-01(config-if)#switchport trunk encapsulation dot1q

vIOS-L2-01(config-if)#switchport mode trunk

vIOS-L2-01(config-if)#switchport trunk native vlan 333

vIOS-L2-01(config-if)#switchport trunk allowed vlan 20,333

vIOS-L2-01(config-if)#no shutdown

vIOS-L2-01(config-if)#exit

vIOS-L2-01(config)#interface GigabitEthernet1/1

vIOS-L2-01(config-if)#switchport trunk encapsulation dot1q

vIOS-L2-01(config-if)#switchport mode trunk

vIOS-L2-01(config-if)#switchport trunk native vlan 333

vIOS-L2-01(config-if)#switchport trunk allowed vlan 20,333

vIOS-L2-01(config-if)#no shutdown

vIOS-L2-01(config-if)#exit

vIOS-L2-01(config)#interface GigabitEthernet1/2

vIOS-L2-01(config-if)#switchport trunk encapsulation dot1q

```
vIOS-L2-01(config-if)#switchport mode trunk
vIOS-L2-01(config-if)#switchport trunk native vlan 333
vIOS-L2-01(config-if)#switchport trunk allowed vlan 20,333
vIOS-L2-01(config-if)#no shutdown
vIOS-L2-01(config-if)#exit
vIOS-L2-01(config)#interface GigabitEthernet1/3
vIOS-L2-01(config-if)#switchport trunk encapsulation dot1q
vIOS-L2-01(config-if)#switchport mode trunk
vIOS-L2-01(config-if)#switchport trunk native vlan 333
vIOS-L2-01(config-if)#switchport trunk allowed vlan 20,333
vIOS-L2-01(config-if)#no shutdown
vIOS-L2-01(config-if)#exit
vIOS-L2-01(config)#vlan 20
vIOS-L2-01(config-vlan)#name VLAN20
vIOS-L2-01(config-vlan)#exit
vIOS-L2-01(config)#vlan 333
vIOS-L2-01(config-vlan)#name VLAN333
vIOS-L2-01(config-vlan)#exit
```

Для остальных надо добавить access на нужные VLAN.

Layer2Switch-3

```
vIOS-L2-01>enable
vIOS-L2-01#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
vIOS-L2-01(config)#interface range GigabitEthernet0/0-3
vIOS-L2-01(config-if-range)#switchport trunk encapsulation dot1q
vIOS-L2-01(config-if-range)#switchport mode trunk
vIOS-L2-01(config-if-range)#switchport trunk native vlan 333
vIOS-L2-01(config-if-range)#switchport trunk allowed vlan 20,333
vIOS-L2-01(config-if-range)#no shutdown
vIOS-L2-01(config-if-range)#exit
vIOS-L2-01(config)#interface range GigabitEthernet1/0-3
vIOS-L2-01(config-if-range)#switchport trunk encapsulation dot1q
vIOS-L2-01(config-if-range)#switchport trunk native vlan 333
vIOS-L2-01(config-if-range)#switchport trunk allowed vlan 20,333
vIOS-L2-01(config-if-range)#no shutdown
vIOS-L2-01(config-if-range)#exit
vIOS-L2-01(config)#no ip domain-lookup
vIOS-L2-01(config-vlan)#name VLAN20
vIOS-L2-01(config-vlan)#exit
vIOS-L2-01(config)#vlan 333
vIOS-L2-01(config-vlan)#name VLAN333
vIOS-L2-01(config)#interface GigabitEthernet0/0
vIOS-L2-01(config-if)#switchport mod access
vIOS-L2-01(config-if)#switchport access vlan 20
vIOS-L2-01(config-if)#no shutdown
vIOS-L2-01(config-if)#exit
vIOS-L2-01(config)#interface GigabitEthernet0/1
vIOS-L2-01(config-if)#switchport mod access
vIOS-L2-01(config-if)#switchport access vlan 333
```

```
vIOS-L2-01(config-if)#no shutdown
vIOS-L2-01(config-if)#exit
```

Layer2Switch-4

```
vIOS-L2-01>enable
vIOS-L2-01#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
vIOS-L2-01(config)#interface range GigabitEthernet0/0-3
vIOS-L2-01(config-if-range)#switchport trunk encapsulation dot1q
vIOS-L2-01(config-if-range)#switchport mode trunk
vIOS-L2-01(config-if-range)#switchport trunk native vlan 333
vIOS-L2-01(config-if-range)#switchport trunk allowed vlan 20,333
vIOS-L2-01(config-if-range)#no shutdown
vIOS-L2-01(config-if-range)#exit
vIOS-L2-01(config)#vlan 20
vIOS-L2-01(config-vlan)#name VLAN20
vIOS-L2-01(config-vlan)#exit
vIOS-L2-01(config)#vlan 333
vIOS-L2-01(config-vlan)#name VLAN333
vIOS-L2-01(config-vlan)#exit
vIOS-L2-01(config)#interface range GigabitEthernet1/0-3
vIOS-L2-01(config-if-range)#switchport trunk encapsulation dot1q
vIOS-L2-01(config-if-range)#switchport mode trunk
vIOS-L2-01(config-if-range)#switchport trunk native vlan 333
vIOS-L2-01(config-if-range)#switchport trunk allowed vlan 20,333
vIOS-L2-01(config-if-range)#no shutdown
vIOS-L2-01(config-if-range)#exit
vIOS-L2-01(config)#interface GigabitEthernet0/0
vIOS-L2-01(config-if)#switchport mod access
vIOS-L2-01(config-if)#switchport access vlan 20
vIOS-L2-01(config-if)#no shutdown
vIOS-L2-01(config-if)#exit
vIOS-L2-01(config)#interface GigabitEthernet0/1
vIOS-L2-01(config-if)#switchport mod access
vIOS-L2-01(config-if)#switchport access vlan 333
vIOS-L2-01(config-if)#no shutdown
vIOS-L2-01(config-if)#exit
vIOS-L2-01(config)#end
```

Layer2Switch-5

```
vIOS-L2-01>enable
vIOS-L2-01#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
vIOS-L2-01(config)#vlan 20
vIOS-L2-01(config-vlan)#name VLAN20
vIOS-L2-01(config-vlan)#exit
vIOS-L2-01(config)#vlan 333
vIOS-L2-01(config-vlan)#name VLAN333
vIOS-L2-01(config-vlan)#exit
vIOS-L2-01(config)#interface range GigabitEthernet0/0-3
```

```
vIOS-L2-01(config-if-range)#switchport trunk encapsulation dot1q
vIOS-L2-01(config-if-range)#switchport mode trunk
vIOS-L2-01(config-if-range)#switchport trunk native vlan 333
vIOS-L2-01(config-if-range)#switchport trunk allowed vlan 20,333
vIOS-L2-01(config-if-range)#no shutdown
vIOS-L2-01(config-if-range)#exit
vIOS-L2-01(config)#interface range GigabitEthernet1/0-3
vIOS-L2-01(config-if-range)#switchport trunk encapsulation dot1q
vIOS-L2-01(config-if-range)#switchport mode trunk
vIOS-L2-01(config-if-range)#switchport trunk native vlan 333
vIOS-L2-01(config-if-range)#switchport trunk allowed vlan 20,333
vIOS-L2-01(config-if-range)#no shutdown
vIOS-L2-01(config-if-range)#exit
vIOS-L2-01(config)#interface GigabitEthernet0/0
vIOS-L2-01(config-if)#switchport mod access
vIOS-L2-01(config-if)#switchport access vlan 20
vIOS-L2-01(config-if)#no shutdown
vIOS-L2-01(config-if)#exit
vIOS-L2-01(config)#interface GigabitEthernet0/1
vIOS-L2-01(config-if)#switchport mod access
vIOS-L2-01(config-if)#switchport access vlan 333
vIOS-L2-01(config-if)#no shutdown
vIOS-L2-01(config-if)#exit
vIOS-L2-01(config)#end
```

Настраиваем адреса на ПК

```
VPCS1
PC1> ip 192.168.20.1/24
VPCS2
PC2> ip 192.168.20.2/24
VPCS3
PC3> ip 192.168.20.3/24
VPCS4
PC4> ip 192.168.20.4/24
VPCS5
PC5> ip 192.168.20.5/24
VPCS6
PC6> ip 192.168.20.6/24
```

2) Результат долгих настроек :D

```
PC1> ping 192.168.20.2

host (192.168.20.2) not reachable

PC1> ping 192.168.20.3

84 bytes from 192.168.20.3 icmp_seq=1 ttl=64 time=8.810 ms
84 bytes from 192.168.20.3 icmp_seq=2 ttl=64 time=7.303 ms
84 bytes from 192.168.20.3 icmp_seq=3 ttl=64 time=8.915 ms
84 bytes from 192.168.20.3 icmp_seq=4 ttl=64 time=6.588 ms
84 bytes from 192.168.20.3 icmp_seq=5 ttl=64 time=6.174 ms

PC1> ping 192.168.20.4

host (192.168.20.4) not reachable

PC1> ping 192.168.20.5

84 bytes from 192.168.20.5 icmp_seq=1 ttl=64 time=7.958 ms
84 bytes from 192.168.20.5 icmp_seq=2 ttl=64 time=3.974 ms
84 bytes from 192.168.20.5 icmp_seq=3 ttl=64 time=7.095 ms
84 bytes from 192.168.20.5 icmp_seq=4 ttl=64 time=5.076 ms
84 bytes from 192.168.20.5 icmp_seq=5 ttl=64 time=8.157 ms

PC1> ping 192.168.20.6

host (192.168.20.6) not reachable

PC1> 
```

```

PC2> ping 192.168.20.1

host (192.168.20.1) not reachable

PC2> ping 192.168.20.3

host (192.168.20.3) not reachable

PC2> ping 192.168.20.4

84 bytes from 192.168.20.4 icmp_seq=1 ttl=64 time=14.760 ms
84 bytes from 192.168.20.4 icmp_seq=2 ttl=64 time=2.790 ms
84 bytes from 192.168.20.4 icmp_seq=3 ttl=64 time=5.031 ms
84 bytes from 192.168.20.4 icmp_seq=4 ttl=64 time=6.649 ms
84 bytes from 192.168.20.4 icmp_seq=5 ttl=64 time=4.278 ms

PC2> ping 192.168.20.5

host (192.168.20.5) not reachable

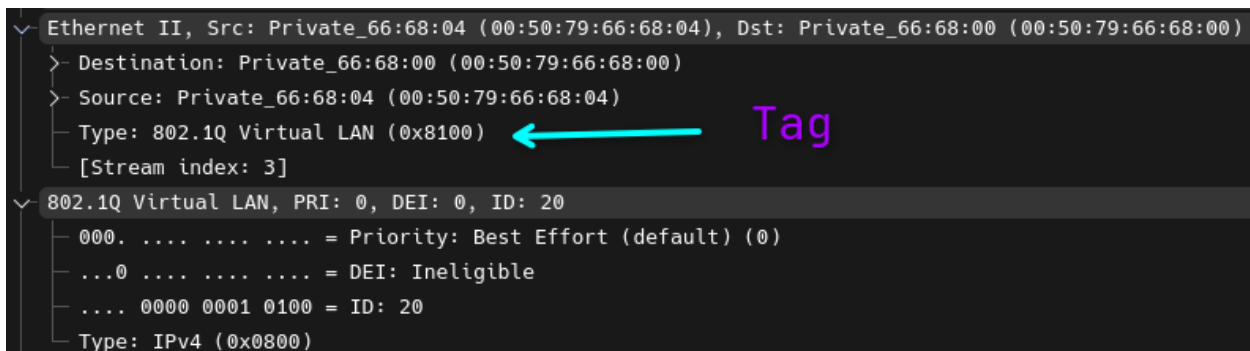
PC2> ping 192.168.20.6

84 bytes from 192.168.20.6 icmp_seq=1 ttl=64 time=8.845 ms
84 bytes from 192.168.20.6 icmp_seq=2 ttl=64 time=2.335 ms
84 bytes from 192.168.20.6 icmp_seq=3 ttl=64 time=8.002 ms
84 bytes from 192.168.20.6 icmp_seq=4 ttl=64 time=3.991 ms
84 bytes from 192.168.20.6 icmp_seq=5 ttl=64 time=9.884 ms

PC2>

```

3) Пингую и смотрю результат через wireshark
для VLAN20



```

Ethernet II, Src: Private_66:68:04 (00:50:79:66:68:04), Dst: Private_66:68:00 (00:50:79:66:68:00)
  > Destination: Private_66:68:00 (00:50:79:66:68:00)
  > Source: Private_66:68:04 (00:50:79:66:68:04)
  | Type: 802.1Q Virtual LAN (0x8100) ← Tag
  | [Stream index: 3]
802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 20
  | 000. .... = Priority: Best Effort (default) (0)
  | ...0 .... = DEI: Ineligible
  | .... 0000 0001 0100 = ID: 20
  | Type: IPv4 (0x0800)

```


Для VLAN333

```
✓ Ethernet II, Src: Private_66:68:05 (00:50:79:66:68:05), Dst: Private_66:68:02 (00:50:79:66:68:02)
  > Destination: Private_66:68:02 (00:50:79:66:68:02)
  > Source: Private_66:68:05 (00:50:79:66:68:05)
  > Type: IPv4 (0x0800) ← No Tag D:
  [Stream index: 3]
✓ Internet Protocol Version 4, Src: 192.168.20.2, Dst: 192.168.20.4
  — 0100 .... = Version: 4
  — .... 0101 = Header Length: 20 bytes (5)
  > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
  — Total Length: 84
  — Identification: 0xa0a9 (41129)
  > 000. .... = Flags: 0x0
  — ...0 0000 0000 0000 = Fragment Offset: 0
  — Time to Live: 64
  — Protocol: ICMP (1)
  — Header Checksum: 0x30a9 [validation disabled]
  — [Header checksum status: Unverified]
```

4) Сохранение файлов конфигурации

```
> tree
.
├── Layer2Switch-1.cfg
├── Layer2Switch-2.cfg
├── Layer2Switch-3.cfg
├── Layer2Switch-4.cfg
├── Layer2Switch-5.cfg
├── module4 lab3.odt
├── PC1_startup.vpc
├── PC2_startup.vpc
├── PC3_startup.vpc
├── PC4_startup.vpc
├── PC5_startup.vpc
└── PC6_startup.vpc

1 directory, 12 files
```